

Ashwin Chembu

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PROFILE

Full-stack and applied-AI engineer with 2+ years of full-time experience building React, TypeScript, Node.js, and retrieval-grounded products from ambiguous customer problems through tested delivery.

TECHNICAL SKILLS

Full Stack: TypeScript, JavaScript, React, Node.js, Express, Python, FastAPI, REST APIs

Applied AI & Data: LLM APIs, RAG, vector search, MongoDB, Snowflake, AWS S3, Redis

Product & Reliability: responsive workflows, pytest, Playwright, UAT, requirements, workflow validation

EXPERIENCE

Business Technology Solutions Associate | ZS Associates

South San Francisco, CA | Jun 2024-Present

Healthcare AI, data, API integration, and workflow delivery

Python | FastAPI | React | Snowflake | AWS | pytest

- Developed Snowflake SQL pipelines for master-data extraction, identifier matching, and API connectivity across **327K+ healthcare records**, producing auditable quality-control reports and exception outputs for repeatable regulated stakeholder review, exception resolution, audit, and stewardship decisions.
- Delivered React and FastAPI knowledge services across **2 evidence paths** - ingested documents and user questions - integrating AWS data services and Redis for grounded enterprise search, auditable document context, and repeatable stakeholder evaluation.
- Converted **3 stakeholder artifacts** - requirements, UAT evidence, and decision logs - into Python/Flask endpoints plus S3 and Smartsheet integrations, connecting product discovery to validated implementation, documented handoffs, and repeatable delivery decisions.

Founder and Lead Developer | Tallyrus

Jun 2023-Present

Full-stack retrieval tooling for educator feedback

Python | React | TypeScript | Flask | MongoDB | LangChain

- Built an AI feedback product across **4 user workflows** using React, TypeScript, Python, LangChain, and Pinecone, translating educator needs into document ingestion, rubric scoring, grounded revision, and review experiences with consistent scoring logic and actionable feedback loops.
- Implemented **5 application safeguards** across the React/TypeScript and Express product, including Google OAuth, protected routes, secure sessions, rate limiting, and Helmet security headers, while MongoDB and S3 managed application and document data for retrieval-based grading workflows.
- Iterated through **3 delivery activities** - user testing, rubric refinement, and full-stack releases - translating educator feedback into scoped product changes and helping raise **five figures** for continued early-stage development and validated review workflows.
- Connected **3 product data systems** - MongoDB application state, S3 documents, and Pinecone retrieval - through Python and Node.js services so rubric edits, uploaded evidence, and grounded feedback remained synchronized across review sessions, scoring decisions, and educator follow-up.

Data Analyst / Web Development Lead | Tech4Good @ UCSC

Mar 2023-Present

Collaborative research and classroom workflow tooling

TypeScript | Node.js | Express | MongoDB | Python | Figma APIs

- Contributed **158 GitHub commits**, including **44 backend-scoped commits** across TypeScript and Figma API chat, polling, Likert, text-entry, and export workflows in a collaborator-owned research toolkit, preserving accurate team ownership, public attribution, and research delivery evidence.
- Implemented **20+ Git-attributed Node/Express REST routes** backed by three MongoDB databases, populated message logs, change streams, and persisted poll and voter records, reliably synchronizing threads, edits, anonymous responses, votes, grouped responses, reveal operations, and live updates.
- Analyzed **43,340 anonymized interaction rows** across **2,805 session IDs** with Python and Pandas notebooks, while six team-published Figma Community resources reached approximately **12.2K aggregate uses** with team-library attribution preserved for reproducible analytics and public research reporting.

PROJECTS

Lume | AI Meeting Intelligence Platform

Dec 2025-Present

Document-grounded meeting knowledge and search

React 18 | Node.js | MongoDB | AWS S3 | WebSockets | RAG

- Built **4 full-stack intelligence workflows** for meeting capture, durable storage, transcript search, and grounded question answering, connecting **React interfaces** with Node.js, MongoDB, AWS S3, and WebSockets for responsive, persistent customer knowledge work.
- Designed a **5-stage retrieval pipeline** covering embeddings, intent routing, evidence extraction, citation-backed generation, and generator/critic refinement so answers remained tied to meeting and document context with visible human-review evidence.
- Unified **2 customer context sources** - transcripts and seed documents - into reusable Project Brain knowledge bases that preserved terminology, relationships, project history, and citation context across repeated search and question-answering sessions for persistent team knowledge.
- Implemented **3 real-time product behaviors** - responsive search, WebSocket updates, and transcript-aware navigation - so meeting capture, evidence review, grounded follow-up, and reusable project context remained connected across active knowledge sessions and responsive customer workflows.
- Separated **2 answer-quality stages** - evidence extraction and generator/critic refinement - from the responsive React experience, preserving citation-backed outputs, visible human-review context, and grounded document evidence before users reused project knowledge across later decisions and sessions.

Sylk | Multi-Tenant Supply-Chain Platform

2025-Present

Role-aware web and Android product with tested tenant isolation

React 19 | TypeScript | Firebase | Capacitor 7 | Playwright

- Built **6 core multi-tenant workflows** covering organization onboarding, shipments, fulfillment, tracking, file collaboration, and real-time chat, separating retailer-only financial data from shared manufacturing views while supporting retailer administrators, factory viewers, and staff across mobile interfaces.
- Enforced permissions for **3 role groups** through Firestore and Storage security rules, separating shared manufacturing workflows from private financial data, then validated cross-tenant access, security rules, and complete end-to-end user journeys with Vitest, **Testing Library**, Firebase emulators, and Playwright.
- Packaged the React product as a **Capacitor 7 Android application**, adding native-safe routing, Firebase Authentication, safe-area handling, legacy WebView fallbacks, mobile file compatibility, and repeatable Gradle workflows that successfully produced an inspected 7.8 MB APK with minimum SDK 23.

Private Filedrop | AI-Integrated File Exchange

2026

Secure APIs and cross-session AI coordination

Node.js | Express | AWS S3 | OpenAPI | MCP | HMAC

- Secured **2 access paths** with password-protected browser uploads and API-key-protected programmatic exchange, adding upload validation, metadata, categories, tags, project paths, auditable version history, and unguessable direct-download links for controlled secure cross-session file handoff systems.
- Implemented **2 storage drivers** - local and S3-compatible - plus documented migration tooling for durable deployment, while REST, OpenAPI, and MCP interfaces exposed upload, download, organization, versioning, message, channel, and webhook operations to browsers, APIs, and AI clients.
- Protected automation using **2 webhook verification controls** with HMAC-SHA256 signatures and timing-safe checks, then added shared mailbox and channel abstractions so sessions could exchange control messages and events with cross-session auditability without exposing storage credentials.

EDUCATION

University of California, Davis | B.S. Computer Science | GPA: 3.7

Davis, CA | Graduated Jun 2025

Relevant Foundation: Algorithms, Databases, Distributed Systems, Operating Systems, AI/ML, Data Structures

PUBLICATION

ICER 2024 - Evaluating Exploratory Reading Groups for Supporting Undergraduate Research Pipelines in Computing - Research examining how structured exploratory reading groups support undergraduate computing research pipelines.

LINKS

Portfolio: <https://ashchembu.com> | LinkedIn: [linkedin.com/in/ashwin-chembu-a5292b205](https://www.linkedin.com/in/ashwin-chembu-a5292b205) | GitHub: github.com/ashwinchembu